

FPGA Synthesis and Prototyping Lab:

41

Counters

The purpose of this experiment is to Synthesis and Check the behavior of the Johnson Counters mentioned in Lab 24

CODE:

```
module jsoncount (
input rst ,
output [3:0] out );

wire clk ;
SB_HFOSC osc_int(
    .CLKHFPU(1'b1),
    .CLKHFEN(1'b1),
    .CLKHF(clk)
);
defparam osc_int.CLKHF_DIV = "0b10"; //12MHZ
reg [22:0] bitcount = 0 ;
reg bitno21 = 0 ;
always @(posedge clk ) begin
    if (!rst ) begin
        bitcount <= 0 ;
        bitno <= 0 ;
        out<= 4'b0000;
        end
    else begin
        bitcount <= bitcount + 1 ;
        bitno <= bitcount [21] ;
        if (bitno == 0 && bitcount[21] == 1'b1 ) begin
            out <= {~out[0], out[3:1]};
        end
    end
end
end
endmodule
```

SYNTH report

```
Top module: \jsoncount

2.43.2. Analyzing design hierarchy..
Top module: \jsoncount
Removed 0 unused modules.

2.44. Printing statistics.

=== jsoncount ===

Number of wires:          13
Number of wire bits:      82
Number of public wires:   6
Number of public wire bits: 31
Number of memories:       0
Number of memory bits:    0
Number of processes:      0
Number of cells:          74
  SB_CARRY                 20
  SB_DFFESR                 5
  SB_DFFSR                 22
  SB_HFOSC                  1
  SB_LUT4                   26

2.45. Executing CHECK pass (checking for obvious problems).
checking module jsoncount..
found and reported 0 problems.

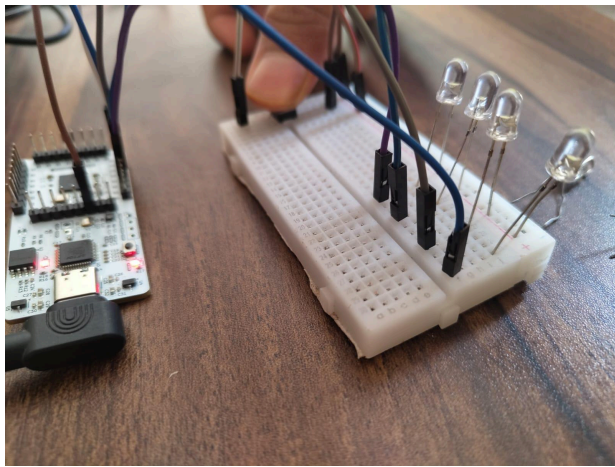
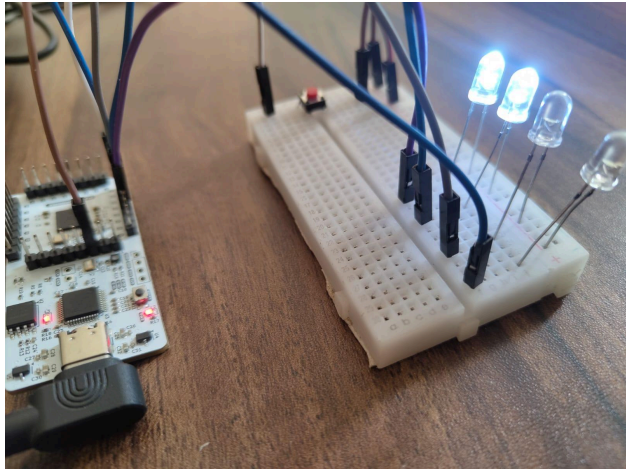
2.46. Executing JSON backend.

Warnings: 5 unique messages, 5 total
End of script. Logfile hash: 3d04a112e7
CPU: user 0.38s system 0.03s, MEM: 45.74 MB total, 40.24 MB resident
Yosys 0.9 (git sha1 1979e0b)
Time spent: 41% 10x read_verilog (0 sec), 13% 1x share (0 sec), ...
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_41$ nextpnr-ice40 --up5k --pack
oncount.asc
Info: constrained 'rst' to bel 'X13/Y31/io0'
Info: constrained 'out[3]' to bel 'X9/Y31/io1'
```

IMPLIMENTATION

```
Info: Program finished normally.
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_41$ icepack jsoncount.asc jsoncount.bin
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_41$ iceprog jsoncount.bin
init..
cdone: high
reset..
cdone: low
flash ID: 0xEF 0x40 0x16 0x00
file size: 104090
erase 64kB sector at 0x000000..
erase 64kB sector at 0x010000..
programming..
done.
reading..
VERIFY OK
cdone: high
Bye.
```

OUTPUTS



OUTPUT VIDEO

https://github.com/KPYerawar/NIELIT-CALICUT-FPGA-PROTOTYPING-ASSIGNMENTS/blob/main/EXP_NO_41/video1.mp4

GITHUB:

https://github.com/KPYerawar/NIELIT-CALICUT-FPGA-PROTOTYPING-ASSIGNMENTS/tree/main/EXP_NO_41